Formal Continuity Without Metaphysical Overreach: What QMT and a `.qbin` Continuity Package Can and Cannot Establish About Identity Across Substrate Transformation

Abstract

This is a work of speculative philosophy and formal methods. It is not an experimental protocol, medical proposal, human-subjects plan, or identity-transfer implementation guide, and it contains no operational procedure for altering, scanning, destroying, or experimenting on any brain or body. Its thesis: Project Nexus's QMT continuity verifier and a versioned `.qbin`-style continuity package can support a rigorous, testable notion of **computational identity continuity for software entities** by preserving constitutionally required state, provenance, and transformation invariants — but that success does not by itself resolve whether analogous preservation would constitute human personal identity, psychological continuity, or phenomenal continuity. Software-agent continuity is the tractable core; the human generalization remains open and, on current evidence, partly beyond any measurement.

1. Why Continuity Is Not One Question

"Continuity" is not one property. At least five distinct questions must be separated: (1) state preservation (are the records restorable?); (2) behavioral preservation (same outputs under the same inputs?); (3) psychological continuity (memories, dispositions, goals, skills, self-model); (4) numerical personal identity (literally the same entity vs. a copy/successor); (5) phenomenal continuity (does first-person experience continue?). These are logically independent: a system can preserve state without behavior, behavior without psychology, psychology without numerical identity (Parfit's branching cases), and — contentiously — all of these while leaving phenomenal continuity undetermined. Nexus's own Identity Architecture already separates identity from state, memory, location, storage realization, similarity, and authorization; the question is how far that sound software separation extends to human persons.

2. Nexus's Existing Digital Continuity Architecture

Per the canonical Nexus 4 and Identity Architecture monographs (reference editions August 2026), the architecture assigns distinct responsibilities: **HVD** (Hyperdimensional Vector Database) is the cognitive database substrate (labeled hypervectors as a direct sum of component spaces); **TMS** (Transdimensional Memory Substrate) owns active state realization, trajectory, copy-on-write structural sharing, and log-structured persistence; **HDF5** is durable long-term persistence reached by staged archival and rehydration, not a hot query backend; **HIGC** defines the constitutional admissible set; **THDE** owns bounded evolution; and **QMT** owns continuity across transformation, migration, rehydration, or representational remapping. The foundational separation is written `identity ≠ state ≠ history ≠ memory ≠ location ≠ relationship`. The

tractable problem is: transformation of a computational entity → preservation of required invariants → provenance/trajectory verification → continuity determination. Crucially, the monograph explicitly lists what Nexus has **not** demonstrated, including "subjective continuity through hardware replacement," "digital immortality," a "philosophical solution to personal identity or machine consciousness," and "phenomenal continuity across restart." Full QMT continuity across arbitrary migrations is itself marked **PROPOSED**, not implemented — the architectural owner is established but the full mechanism is not built. Prior verified Nexus work includes a quinary hypervector substrate, HTNLA binding, deterministic k-WTA, and an 11-dimensional RK4 controller executing correctly and passing invariant/determinism checks; these are engineering results, not evidence about consciousness. mcp__Google_Drive__read_file_content + 4

Nexus documents are cited here only to state what Nexus claims and has built. They are never used as independent proof of any external claim in neuroscience, physics, or philosophy of mind.

3. `.qbin`: Serialization Is Not Identity

The inspected Nexus source (`qbin.py`, HVD 1.5) implements `.qbin` as a streaming container: an 8-byte magic `HVDQBIN1`, a big-endian manifest length, a serialized `QBinManifest` (Protobuf), and concatenated artifact payloads, each carrying a BLAKE2b-256 digest. The code comment is explicit: "A qbin does not replace HDF5. It packages the canonical HDF5 memory archive with a Protobuf manifest and optional supplemental artifacts so a complete HVD state can be moved between substrates." The digests "provide corruption integrity, not authenticity; signing is a separate trust concern." A test suite (`test_qbin_v15.py`) exercises packaging an HDF5 archive, inspecting it, extracting artifacts, and "rebirthing" into a fresh HVD, plus a corruption-detection test that flips one byte and asserts an "artifact digest mismatch." This is a real, working serialization-and-rehydration mechanism. PhilArchive ↗ Isaak ↗

We must sharply distinguish three properties that `.qbin` bears on: **serialization completeness** (are all required descriptors captured?), **functional reproducibility** (does re-execution reproduce behavior?), and **identity continuity** (is it the same logical entity?). These are not equivalent: complete serialization does not guarantee successful re-execution (missing dependencies, nondeterminism); successful re-execution does not guarantee preserved logical continuity (an independently instantiated identical copy re-executes perfectly yet is a distinct entity); and preserved software-agent continuity does not demonstrate phenomenal continuity. A `.qbin` package that would support continuity (not mere backup) must plausibly preserve: logical entity identifier; schema/version; constitutional baseline/identity constraints; HVD logical-state descriptors; TMS generation/durable-state references; HTLA label schema; HTNLA parameters/activation definitions; model/version identifiers; deterministic replay state; event-ledger/trajectory tip; provenance hashes; a QMT transformation manifest; source/destination generation identifiers; representation transformations; numerical precision/encoding; random-state/seed material; cryptographic integrity data; required external dependencies; explicitly typed unavailable state; and reconstruction evidence. Some are **required** (logical ID, schema/version, provenance hashes, trajectory tip, integrity data), some **optional** (supplemental artifacts), and some **architecture-specific** (quinary/HSL encoding, HTNLA activation params). Authorization metadata may be *carried* but must never be treated as

identity — see §16. The current implemented `.qbin` is closest to a **manifest-plus-referenced-state** package over content-addressed (digest-bearing) payloads; self-contained, incremental, and fully content-addressed variants are design options with different trade-offs (portability vs. size vs. dedup vs. history).

4. Formalizing QMT for Software Entities

For a transformation T: S_a → S_b along a trajectory τ = (S_a, e_1, S_1, …, e_n, S_b) with invariant set I = {I_1,…,I_m}, a candidate continuity predicate is:

Continuity_QMT(S_a,S_b,τ,T,I) ⇔ ValidTrajectory(τ) ∧ AuthorizedTransformation(T) ∧ ProvenanceClosure(τ) ∧ PreservedRequiredInvariants(I) ∧ LogicalStateCompatibility(S_a,S_b) ∧ NoUnresolvedFork(τ).

This formulation is defensible but should be adopted critically. Its components map onto established formal concepts, and QMT's contribution is best understood as their *assembly* around a constitutional invariant set rather than a new equivalence. `LogicalStateCompatibility` is a **decoding/canonicalization** relation (state-equivalence up to representation), close to refinement mappings and data-refinement in state-machine theory. `PreservedRequiredInvariants` is a **safety-invariant** condition familiar from temporal-logic verification and replicated state machines. `ProvenanceClosure` corresponds to **cryptographic provenance / event-sourced lineage** (hash-chained ledgers, Merkle DAGs, content-addressed object stores as in Git/IPFS). `ValidTrajectory` + `NoUnresolvedFork` correspond to **single-writer / non-equivocation** guarantees and split-brain avoidance in distributed systems. `AuthorizedTransformation` is a *policy* predicate, not a behavioral one.

What QMT adds beyond ordinary state equivalence is the insistence that continuity is a **path property with provenance and non-forking authority**, not snapshot equality and not behavioral indistinguishability. This is exactly the gap that pure bisimulation misses: two processes can be bisimilar (or trace/observationally equivalent) yet be distinct entities with distinct histories. Conversely, checkpoint/restart, deterministic replay, and live VM migration (vMotion-style: pre-copy memory, suspend, transfer non-memory state, resume) already treat "same running entity across physical relocation" as routine — the Identity Architecture migration sketch (quiesce → signed continuity manifest → transfer → rehydrate → QMT verify → atomically promote) is a direct analogue. Concepts that fit QMT: refinement mappings, safety invariants, event-sourced provenance, replicated-state-machine single-writer authority, checkpoint/restart. Concepts that fit only partially: bisimulation/observational equivalence (necessary for behavioral level, insufficient for identity); process calculi congruences (about behavior, not lineage). The monograph correctly notes continuity may be a **directed reachability relation** (x_a ⇝_I x_b) rather than a symmetric equivalence, because key rotation, irreversible remediation, or schema migration can preserve identity without being reversible.
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5. Four Levels of Preservation

(A) Bitwise. Byte-for-byte equality. Too strong and usually unnecessary; almost every legitimate operation changes bytes.
(B) State-equivalent. Different encoding, equivalent logical state; formalizable via decoding maps / canonicalization. This is the natural target for `.qbin` migration.

(C) Functional/behavioral. Observationally equivalent under a specified test domain; formalized by bisimulation, trace/observational equivalence, refinement, conformance testing.
(D) Identity continuity. Same historical/constitutional entity through a valid transformation trajectory: provenance, non-forking authority, permitted transformation, preserved constitutional invariants, causal trajectory, authorized promotion, QMT validation.

The decisive point for the human case: **(C) does not entail (D).** Two independently instantiated copies can be behaviorally identical (perfect bisimulation) while being numerically distinct logical entities with different provenance. Behavioral equivalence is symmetric and history-blind; identity continuity is historical and, in general, directed. This is the formal seed of the branching problem.

6. The Copy/Fork Problem

Let D_0 branch at time t into perfect copies D_A and D_B. Formally both can be state-equivalent (B) and behaviorally equivalent (C) to D_0 at the instant of branching. But Nexus logical identity cannot coherently grant *numerical* identity to both without contradiction: identity as a path property with non-forking authority forces the post-branch relation to become **lineage-specific**. After branching, "same identity" degrades from an equivalence relation to a **provenance-indexed relation**; provenance *can* distinguish otherwise-identical descendants (distinct trajectory tips, distinct authorized-promotion events), which is precisely why the candidate predicate includes `NoUnresolvedFork`. Distributed-systems reasoning supplies the governing intuition: replica count ≠ identity count; concurrent internally-valid extensions of a common tip (H_k → H_{k+1}^A and H_k → H_{k+1}^B) are a split-brain that must be classified (benign concurrency, causal divergence, stale extension, equivocation, or unresolved) rather than silently merged, and non-mergeable transitions require a single mutation authority (lease/quorum/fencing).
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Parfit's branching analysis maps onto this directly and is the bridge to the human case: he argues that in fission, *identity is not what matters* — what matters is Relation R (psychological connectedness/continuity with the right cause), which can hold one-to-many. For software, Nexus can *stipulate* a non-forking authority rule so that at most one branch inherits the production identity while others become explicitly derived/sandbox entities. For persons, no such stipulating authority exists, and the intuition that "I cannot be two people" collides with the symmetry of a perfect fission. The key difference: software identity can be **defined** by an authority layer with a canonical ledger; human personal identity cannot be legislated into determinacy by fiat, which is why the same formal machinery leaves the human question open.

7. Five Questions About Human Uploading (Chalmers Decomposition)

Chalmers separates (and we must never collapse into a single "fidelity" score): behavioral continuity; psychological continuity; consciousness; numerical identity; and survival value ("as good as survival" even absent strict numerical identity). His organizational-invariance principle, defended via the fading-qualia and dancing-qualia arguments, gives a reason to think a fine-grained functional isomorph *could* be conscious — but Chalmers is explicit that this does not settle identity: a functional isomorph (a molecule-for-molecule twin) is qualitatively identical yet numerically distinct — "it is not me," and "if you kill me… and preserve the

twin, I will die." Thus organizational invariance, even if true, secures at most consciousness and behavioral/psychological continuity, not numerical identity or guaranteed survival. PhilPapers + 2 ↗

Question	Formally specifiable?	Behaviorally measurable?	Physically measurable?	Metaphysically decidable?	Current Nexus relevance
Behavioral continuity	Yes (bisimulation, conformance)	Yes	Yes (I/O tests)	N/A	High — QMT/`.qbin` directly address it for software
Psychological continuity	Partly (state/memory schemas)	Partly (memory/skill probes)	Indirectly	Contested	Moderate — HVD/TMS preserve memory-analog state
Consciousness	No accepted spec	No (report ≠ presence)	No accepted measure	Disputed (may be intractable)	None — explicitly disclaimed
Numerical identity	Only relative to a stipulated authority	No	No	Disputed / possibly no fact	Software: yes by fiat; human: open
Survival value	No (normative)	No	No	Normative/contested	None

8. Personal Identity in Philosophy

Expert opinion is genuinely and persistently divided. In the 2020 PhilPapers survey (Bourget & Chalmers; 1,615 respondents on this question), personal identity drew 43.65% for the psychological view (39.44% exclusive), 19.07% biological (15.60%), and 14.86% further-fact (13.37%), with the remainder split across "no fact of the matter," "too unclear," alternative views, and agnostic. On the teletransporter (new matter), respondents split 351 vs. 238 (accept/reject) on survival among psychological-view holders; overall roughly evenly divided. On "mind uploading (brain replaced by digital emulation)," 27.5% (n=279) answered survival vs. 54.2% (n=551) death, with 18.4% "other." Parfit's *Reasons and Persons* (1984) supplies the load-bearing argument that identity may not be what matters and that in some cases identity questions are "empty." Chalmers's *Mind Uploading: A Philosophical Analysis* and "The Singularity: A Philosophical Analysis" (2010) supply the decomposition and the optimistic/pessimistic framing. This distribution is itself evidence: the numerical-identity and survival questions are not settled by argument, and the paper does not pretend otherwise. philpeople + 2 ↗

9. Gradual Replacement vs. Scan-and-Copy (Analysis Only)

Chalmers's fading-qualia and dancing-qualia arguments (in "Absent Qualia, Fading Qualia, Dancing Qualia," 1995) target gradual functional substitution: if functional organization is preserved at a fine enough grain, consciousness should neither fade undetectably nor dance without notice, on pain of implausible dissociations between experience and self-report. Wiley & Koene ("The Fallacy of Favoring Gradual Replacement Mind Uploading Over Scan-and-Copy," J. Consciousness Studies 23(3–4):212–235, 2016) argue there is no principled replacement-rate cutoff distinguishing survival from non-survival, challenging the intuition that gradual is "safer" than scan-and-copy. Conceptually — and never as a procedure — QMT-style *trajectory* continuity maps more naturally onto **gradual in-place transformation** than onto destructive scan→reconstruction, because trajectory continuity is defined by an unbroken provenance path with preserved invariants, which a gradual causal process supplies and a discontinuous copy does not. But this is an analogy about the *form* of the continuity relation, not evidence that either route preserves phenomenal continuity in humans. Biological/embodied objections (below) press that even fine-grained functional organization may omit consciousness-relevant substrate. PhilPapers ↗ arXiv ↗

10. Whole-Brain Emulation: Actual State of the Field Through 2026

Sandberg & Bostrom's *Whole Brain Emulation: A Roadmap* (Technical Report #2008-3, Future of Humanity Institute, Oxford) remains the structural framing: scanning, translation to functional models, and simulation. Sandberg and Bostrom (2008) estimate approximately 50 percent probability of WBE by 2064, "though timelines depend critically on which neural features prove computationally essential" — a conditional 2008 projection, not a scheduled date. The 2025 *State of Brain Emulation Report* (Zanichelli, Schons, Freeman, Shiu, Arkhipov; arXiv:2510.15745, Sept 2025) documents real progress and hard limits. On recording: "Electrophysiology has moved from a few hundred simultaneously recorded neurons to several thousand using high-density probes… Voltage imaging, capable of resolving single spikes, has progressed from hundreds of neurons to tens of thousands," an aggregate rise of roughly two orders of magnitude in effective data bandwidth since 2008. Calcium imaging "now captures… about a million cortical neurons (~1.4% of total neurons) in mice." On structure: multiple complete adult *Drosophila* connectomes (FlyWire, 2024), with cost per reconstructed neuron falling "from an estimated $16,500 in the original *C. elegans* connectome to roughly $100 in recent larval zebrafish projects." But whole-brain single-neuron single-spike coverage remains unattained in any organism: the report states "neuroscience has not yet achieved whole-brain recording (≥ 95% of neurons and brain volume) at single-neuron resolution in any organism. The closest achievements include larval zebrafish with approximately 80% brain coverage" (and ~50% of neurons in *C. elegans*). Further, structure–function datasets from the *same individual* are exceptionally rare; connectomes largely lack molecular annotation; neuromodulation is largely omitted from models; and a central unresolved question is explicitly "which biological details are necessary for a faithful emulation." The report deliberately does not address consciousness or personality preservation. arxiv ↗

Layer	Model organisms (2026)	Human status
Structural connectome (synaptic)	*C. elegans*, adult *Drosophila* complete; zebrafish underway	Only a petavoxel cortical *fragment* reconstructed (Shapson-Coe et al., Science 2024)
Functional dynamics	~1M mouse neurons (~1.4% of total, calcium imaging); ~80% zebrafish coverage	Whole-brain single-neuron single-spike: not achieved (no organism ≥95%)
Cellular/synaptic state	Partial	Not achieved
Molecular/neuromodulatory state	Sparse	Not achieved
Developmental/embodied history	Partial in worm/fly	Not achieved

Human whole-brain emulation is far beyond demonstrated capability; the honest statement is that required biological detail remains genuinely uncertain.

11. What Brillouin Microscopy Actually Measures

Brillouin light-scattering microscopy measures a **Brillouin frequency shift and linewidth** of GHz acoustic phonons, which — given assumed refractive index and mass density — yield the complex longitudinal (viscoelastic) modulus (storage modulus M′ and loss modulus/longitudinal viscosity). It is non-contact, label-free, and diffraction-limited. This is established by the consensus statement of ~40 authors (Bouvet, Prevedel, Scarcelli, Kabakova, Palombo, Elsayad et al.), "Consensus Statement on Brillouin Light Scattering Microscopy of Biological Materials," published in *Nature Photonics* 19:681–691 (2025), DOI 10.1038/s41566-025-01681-6, and the review by Kabakova et al., "Brillouin microscopy," *Nature Reviews Methods Primers* 4:8 (2024). Validated/leading biomedical uses: corneal biomechanics and keratoconus (the most clinically mature; motion-tracking Brillouin, *Ophthalmology* 131(3):310–321, 2024); live embryo development (Bevilacqua et al., *Nature Methods* 20:755–760, 2023); tumor tissue mechanics; and ex-vivo amyloid-plaque viscoelasticity (Mattana et al., correlative Brillouin–Raman in transgenic mouse brain). It measures mechanical/viscoelastic (and refractive-index/density-dependent) properties — not electrical, chemical-identity, or informational content. **No established scientific evidence currently connects Brillouin microscopy measurements to personal identity or phenomenal continuity**, nor to memory content, subjective states, or any "resonance signature" of a self. This holds regardless of any historical assumption to the contrary. Nature ↗

12. What NV-Diamond Magnetometry Actually Measures

Nitrogen-vacancy (NV) diamond magnetometry measures the **time-varying magnetic field produced by ionic currents of neuronal action potentials**. The foundational demonstration is Barry, Turner, Schloss,

Glenn, Song, Lukin, Park & Walsworth, "Optical magnetic detection of single-neuron action potentials using quantum defects in diamond," *PNAS* 113(49):14133–14138 (2016), DOI 10.1073/pnas.1601513113: single-neuron sensitivity, room-temperature/ambient operation, demonstrated on excised marine-worm (*Myxicola infundibulum*) and squid giant axons and exterior to intact living marine worm, noninvasively and label-free. The diamond sensor "is biocompatible and can be brought into close proximity to the organism without adverse effect, allowing for long-term observation," because "the light that probes the NV sensors stays within the biocompatible diamond chip and does not enter the organism" (sensor ~10 μm proximity). As of 2026, NV single-neuron AP sensing remains **invertebrate / in-vitro / ex-vivo**, with the sensor outside the tissue; mammalian NV work is tissue-slice compound-action-potential level (e.g., *Sci. Rep.* 2023) or simulation only (Karin et al., modeling a ~36 pT mammalian AP field near current NV limits). The only in-vivo mammalian single-neuron *magnetic* AP recording to date used a different technology — giant-magnetoresistance sensors (Klein et al., *J. Neurophysiology*, 2024) — not NV diamond. The three claims "detects a magnetic field," "detects cognition," and "detects identity" are separate and require separate evidence. **No peer-reviewed literature establishes an identity-, consciousness-, or memory-specific magnetic signature** detectable by NV magnetometry or any AP magnetic method. DSpace@MIT ↗ arxiv ↗

Property	Brillouin microscopy	NV-diamond magnetometry
Physical quantity	GHz phonon frequency shift/linewidth → longitudinal viscoelastic modulus	Magnetic field of action-potential currents
Spatial scale	Sub-micron, 3D, optical diffraction limit	Single-neuron; sensor ~10 μm proximity
Invasiveness	Non-contact, label-free	Noninvasive, label-free; sensor external
Demonstrated bio use	Cornea, embryo, tumor, plaque viscoelasticity	Invertebrate/excised neurons; intact marine worm
In-vivo mammalian single-neuron?	N/A	Not achieved (NV); GMR only
Identity/consciousness signature?	None in literature	None in literature

13. Why QHRF Does Not Currently Establish Identity

"QHRF" as a purported joint measurement establishing a latent identity/consciousness state is **not** an established observable in the scientific literature. The acronym appears in unrelated, speculative, or non-peer-reviewed contexts (e.g., a "Quantum Harmonic Resonance Framework" for entangled-photon

authentication; quantum fingerprinting for device localization), none of which measure personal identity or phenomenal continuity. There is no calibration target, no validated observable corresponding to "identity preservation," and — decisively — no justified generative/likelihood model connecting any Brillouin or NV measurement distribution to an identity hypothesis. This is a formal, not merely empirical, obstacle: a Bayesian posterior P(identity preserved | measurement) is only as meaningful as the likelihood P(measurement | identity preserved) that links data to hypothesis. Absent an established measurement model, priors and mathematical machinery cannot manufacture empirical identification; a precise-looking posterior would be a category error. QHRF is therefore classified here as a **HISTORICAL/SPECULATIVE CONSTRUCT**. Academia.edu ↗

Excluded historical material. Nexus-adjacent draft materials (a "deep research prompt" file and its antecedents) are recorded as having contained an operational identity-transfer protocol: a Bayesian destructive-transfer threshold (a numeric posterior gate for proceeding), "authorized_to_proceed" decision logic, an IRB/consent checklist, QHRF instrumentation specifics presented as an acquisition plan, protocol-specific data schemas, and a Composite Identity Fidelity pass/fail metric. Per the hard boundary of this investigation, that material existed, is explicitly rejected/superseded, and is excluded in full; none of it is reproduced, summarized in operational detail, or reconstructed here. Its removal is a requirement of rigor, not a limitation of scope: no responsible treatment of this topic at the current state of the science produces an operational protocol, and no numeric go/no-go threshold for an irreversible act can be justified without a measurement model that does not exist.

14. What Could Ever Be Measured?

Evidence track	Empirically testable now?	Testable in principle?	May remain metaphysical?
(A) Software continuity	Yes (QMT proof obligations, replay, migration)	Yes	No
(B) Functional human emulation	No (far beyond capability)	Plausibly, if WBE matures	No, if organizational invariance holds
(C) Psychological continuity	Partly (memory/skill/behavior probes)	More fully with better instruments	Contested at the margins
(D) Numerical personal identity	No	Possibly never (may be no fact)	Yes
(E) Phenomenal continuity	No accepted external measure	Unknown; may be intractable	Yes

The five tracks must never be collapsed into one score. Track (A) is where QMT has genuine, testable value. Tracks (D) and (E) may not reduce to measurement even with arbitrarily advanced instruments.

15. What QMT Can Legitimately Claim

Under stated assumptions, QMT and a `.qbin`-style package **can support**: (i) a formal, checkable notion of software-agent continuity across serialization, migration, rehydration, compaction, and representational remapping; (ii) detection of corruption and unresolved forks; (iii) provenance closure via hash-chained lineage; (iv) preservation of declared constitutional invariants across directed (possibly irreversible) transformations; and (v) a principled separation of continuity from mere similarity, mere behavioral equivalence, and mere byte equality. These are engineering claims with clear falsification conditions (drift experiments, fork/rollback adversaries, replay determinism, composition of continuity across transformation chains).

16. What QMT Cannot Legitimately Claim

QMT and `.qbin` **cannot** establish: (i) that behavioral or state equivalence constitutes numerical identity (two copies refute this); (ii) that software continuity implies human psychological, numerical, or phenomenal continuity; (iii) that successful snapshot restoration preserves consciousness; (iv) that deterministic computation preserves personal identity; or (v) that any instrument (Brillouin, NV, or a composite) measures identity or phenomenal continuity. Continuity also must not be conflated with **authorization**: as the monograph states, `SameIdentity ⇏ Authorized`. A restored agent should not inherit every capability, credential, or external authority from identity continuity alone — the correct analogues are credential rotation, certificate/key rotation with staged trust overlap, and capability-scoped authorization, which keep authentication, authorization, and capability restoration as separate predicates from continuity proof.
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17. Open Research Questions (No Milestones, No Procedures)

Can QMT's invariant set be specified concretely enough to admit a category-theoretic or refinement-based composition theorem (does continuity compose across transformation chains)?
What is the minimal `.qbin` content sufficient for continuity vs. mere reproducibility, and can "required vs. optional vs. architecture-specific" be proven rather than asserted?
Can non-forking authority be made robust under distribution (fork classification, fencing) without circularly assuming the identity it verifies?
Which biological details are necessary for faithful emulation (the WBE field's own central open question)?
Is there any route to an external measurement model for psychological continuity that does not smuggle in unvalidated identity assumptions?
For phenomenal continuity: is there, even in principle, an external observable — or is this permanently first-person?

18. Adversarial Self-Review

The strongest objections to the paper's optimistic reading of the human generalization, presented fairly:

Biological naturalism / biological computationalism (Searle; Seth 2025; Cao 2022): functional properties in the brain are not nomologically independent of material properties (metabolism, neurochemistry are intertwined with computation), so functional/organizational preservation may omit consciousness-relevant substrate. If true, WBE sufficiency and substrate independence fail, and Track (B)→(E) inference collapses.
Anti-functionalist critiques of organizational invariance: fading/dancing-qualia arguments have been challenged as question-begging (assuming the organizational-invariance conclusion at the neuron/silicon grain) and as underrating the possibility of introspectively inaccessible phenomenal change.
Connectome-insufficiency: infrequent strong connections and molecular/neuromodulatory state constrain connectomic prediction of function (Cell 2025); a wiring diagram underdetermines dynamics.
Copy/branching against psychological survival: Parfit's own conclusion that identity "is not what matters" cuts both ways — it may mean uploads are "as good as survival," or that nothing that matters survives a copy. Eskildsen Group ↗
The abstract-object challenge (Zhan 2024, *Synthese*): if uploading requires minds to be abstract objects (multiply instantiable), it is incompatible with leading concrete-particular theories of personal identity.

What would falsify or materially weaken the central thesis? The *software* thesis (Track A) would be weakened if: continuity cannot be preserved across ordinary storage transformation without freezing physical representation; if QMT cannot define transformation continuity without circularly assuming identity; or if fork ambiguity cannot be made explicit and fail-closed at scale. The *human-generalization boundary* (the claim that software success does not settle the human case) would be *overturned* — i.e., the human case would become tractable — only if a validated measurement model linking a physical observable to phenomenal/numerical continuity were established; on current evidence (Brillouin, NV, WBE state-of-field) no such model exists, so the boundary stands. Conversely, the thesis would be *strengthened* wherever behavioral/psychological continuity proves measurable while numerical/phenomenal continuity remains disputed — which is what the PhilPapers distribution and the instrument audits show.

19. Claim-Status Audit

Claim	Status
Identity ≠ state ≠ memory ≠ location ≠ similarity ≠ authorization (software)	CANONICAL
`.qbin` streaming container with Protobuf manifest + BLAKE2b digests	IMPLEMENTED
`.qbin` package/inspect/extract/rebirth + corruption detection	VERIFIED (test artifact)
Quinary hypervector substrate, HTNLA binding, deterministic k-WTA, 11D RK4 controller	VERIFIED (prior Nexus work)

Claim	Status
Identity Event Ledger (append-only, SHA-256 chained), replayable active-TMS, Phase 3B HIGC, Phase 3C THDE	IMPLEMENTED
QMT continuity across arbitrary migration/rehydration	PROPOSED (owner established, mechanism not built)
Learned local tolerance; autonomous host mutation	PROPOSED / NOT AUTHORIZED
Candidate continuity predicate Continuity_QMT(...)	INFERRED (synthesized formalization)
Organizational invariance secures consciousness in isomorphs	RESEARCH HYPOTHESIS (contested)
Whole human brain emulation	DESIGN TARGET / far beyond current capability
Brillouin → viscoelastic modulus	EXPERIMENTAL (well-validated for mechanics)
NV magnetometry → single-neuron AP magnetic field (invertebrate/ex-vivo)	EXPERIMENTAL
QHRF as identity observable	HISTORICAL/SPECULATIVE (rejected)
Bayesian destructive-transfer threshold, authorized_to_proceed, IRB/consent, Composite Identity Fidelity metric	HISTORICAL/SUPERSEDED (excluded, not reconstructed)
Software continuity ⇒ human phenomenal continuity	NOT ESTABLISHED (category boundary)

Diagrams (described)

Software case (continuity holds): HVD logical identity → active TMS state → coherent generation → `.qbin` package (manifest + digested artifacts) → transformation/migration → rehydration/reconstruction → QMT verification → PASS → *the same logical entity continues.*

Human thought experiment (unresolved): biological brain → hypothetical functional capture → reconstruction → behavioral resemblance? → memory continuity? → conscious? → numerically the same person? → as-good-as-survival? — *every one of these terminal nodes remains an open question mark; the diagram deliberately does not resolve them.*

Responsibility separation: TMS (live active memory/trajectory) — HDF5 (durable long-term persistence) — HVD (logical cognitive database) — QMT (continuity verifier across transformation). These are distinct responsibilities and must not be collapsed: durable storage is not continuity; a database handle is not a self; a fast projection is not authority.

Conclusion

The central thesis survives testing, in narrowed form: QMT and a `.qbin`-style continuity package can support a rigorous, falsifiable notion of computational identity continuity for software entities by preserving constitutionally required state, provenance, and transformation invariants — and that success, real as it is, does not by itself resolve whether analogous preservation would constitute human personal identity or phenomenal continuity. The software problem is tractable because identity can be *defined* by a constitutional authority over a canonical ledger; the human problem is not, because numerical and phenomenal continuity are, on current evidence, either philosophically disputed or beyond any established measurement. The honest position is neither hype nor dismissal: build and verify the software continuity architecture on its own terms, and hold the human questions open.

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